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Original article

The influence of training task stimuli on transfer effects of working memory training in aging^{☆,☆☆}

L'influence des stimuli de la tâche d'entraînement de la mémoire de travail sur les effets de transfert dans le vieillissement

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ABSTRACT

Introduction. – Working memory (WM) training is known to produce benefits in older adults' WM. Transfer effects to untrained abilities, however, remain controversial and several aspects are thought to influence the generalization of benefits, including the kind of stimuli used in the training tasks, an aspect rarely addressed in older adults.

Objective. – The present study had two aims: (1) to test the efficacy of a visuospatial WM training procedure in older adults, in terms of specific and transfer effects; (2) to examine in two experiments whether the type of stimuli used in the training task influences the training's effectiveness. Experiment 1 adopted images with a

[☆] Both the neutral and the emotionally positive images used in the criterion tasks (in both experiments) were drawn from the *International affective picture system* (IAPS) database; (Lang, Bradley, & Cuthbert, 1998). The images were chosen on the basis of two important criteria: (1) neutral valence; (2) high visibility. Given the use of the images within the matrices, only stimuli that were easily identified were used, avoiding the more complex images (the size of the stimulus in the matrix was 2.3 × 2.3 cm).

^{☆☆} For Experiment 1, the mean values were 5.15 (SD=0.43) for emotional valence, and 3.53 (SD=0.83) for arousal. In Experiment 2, they were 7.20 (SD=0.47) for valence and 5.27 (SD=0.76) for arousal.

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neutral valence while experiment 2 used emotionally positive images based on evidence that older adults tend to remember positive stimuli better. In both experiments, specific training-related gains in a visuospatial WM task (the criterion task) and transfer effects on measures of verbal WM, visuospatial short-term memory, processing speed and reasoning were examined. Maintenance of training benefits was also assessed at an 8-month follow-up.

Method. – Seventy older adult (63–75 years old) volunteers (35 for experiment 1, and 35 for experiment 2) were randomly assigned to a training or active control group. The same visuospatial WM training procedure was used in both experiments, manipulating only the type of stimuli used (neutral in experiment 1 and emotionally positive in experiment 2).

Results. – In both experiments, only trained participants showed specific benefits in the WM criterion task. These gains were also maintained at the follow-up, but no transfer effects were identified.

Conclusion. – Overall, our findings using the present visuospatial WM training paradigm suggest that it is less effective, in terms of transfer effects, than the same paradigm administered verbally in a previous study, regardless of the type of stimuli used in WM training tasks (neutral or emotionally positive stimuli).

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R É S U M É

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Entraînement de la mémoire de travail
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S'il a été montré que l'entraînement de la mémoire de travail (MdT) produit des améliorations de la MdT chez les personnes âgées, les effets de transfert de tels entraînements restent controversés. L'étude présentée a deux objectifs : tester l'efficacité d'une procédure d'entraînement de la MdT visuospatiale et examiner l'effet du type de stimuli utilisés dans la tâche d'entraînement sur le résultat de l'entraînement – en termes d'effets spécifiques et de généralisation à des tâches non entraînées. Soixante adultes âgés (63–75 ans) ont suivi une même tâche d'entraînement, comportant soit des stimuli neutres (expérience 1), soit à valence positive (expérience 2). Les résultats montrent un effet positif de l'entraînement sur la tâche critère de la MdT (également maintenu après 8 mois) mais aucun effet de transfert n'a été mis en évidence, indépendamment du type de stimuli utilisés dans les tâches d'entraînement visuospatiale de la MdT.

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1. Introduction

Working memory (WM), or the ability to retain and simultaneously process information in order to execute cognitive tasks (Badddeley & Hitch, 1974), undergoes a linear decline with aging (e.g., Park et al., 2002). Thus, how to effectively enhance WM performance in older adults has become a topic of particular interest due to the strong involvement of WM in higher-order cognition, and the important consequences of its decline in terms of daily life skills (Willis & Marsiske, 1993).

Promising results have been obtained with process-based WM training, which targets general WM mechanisms (i.e., executive attention) and involves repetitive and intensive practice with WM tasks. Moreover, as shown by a recent meta-analysis (Karback & Verhaeghen, 2014), WM training

in older adults produces large specific effects as well as transfer ones within the same or nearby domains (the so-called nearest or near transfer effects, respectively). Far transfer effects (benefits in abilities less directly related to WM) are also present, although they are reduced in terms of effect size when compared to specific effects and near transfer effects (ranging about 0.4 SD). As WM training procedures differ in many aspects, such as duration and intensity, participants' age, and the use of active versus passive control groups, the crucial factors mediating training efficacy in older adults are not clear yet. It must be noted, however, that the presence and the extent of the transfer effects remains, in general, controversial and has sparked debate on the efficacy of WM training (e.g., [Karback & Verhaeghen, 2014](#)).

One important aspect, rarely taken into consideration, that may contribute to divergences in results from WM training studies' in older adults, concerns the modality (verbal or visuospatial) used for the WM training tasks. Among about 16 WM training studies conducted on older adults, only 7 (to our knowledge) adopted a verbal modality (i.e., [Dahlin, Neely, Larsson, Bäckman, & Nyberg, 2008](#); [Borella, Carretti, Riboldi, & De Beni, 2010](#); [Borella, Carretti, Zanon, Zavagnin, & De Beni, 2013](#); [Carretti, Borella, Fostinelli, & Zavagnin, 2013](#); [Stepankova et al., 2014](#); [Cantarella, Borella, Carretti, Kliegel, & De Beni, in press](#)), while the majority adopted a visuospatial modality (e.g. [Li et al., 2008](#); [Buschkuhl et al., 2008](#); [Heinzel et al., 2014](#); [Borella et al., 2014](#)), or both modalities ([Richmond, Morrison, Chein, & Olson, 2011](#); [Brehmer, Westerberg, & Bäckman, 2012](#); [Zinke, Zeintl, Eschen, Herzog, & Kliegel, 2012](#); [von Bastian & Oberauer, 2013](#); [Zinke et al., 2014](#)). Recent meta-analyses reached contrasting conclusions concerning the impact of training modality on training efficacy. In one of these meta-analyses, modality was not a significant moderator of any transfer effects ([Schwaighofer, Fischera, & Bühner, 2015](#)), while in another, the modality and in particular, visuospatial WM, was found to lead to more persistent training and near transfer gains than verbal WM training ([Melby-Lervåg & Hulme, 2013](#)). Nonetheless, these studies included various age groups and many different types of training task (verbal, visuospatial, or dual) neither of them focused specifically on older adults.

Important food for thought in literature on aging comes from the study of [Borella et al. \(2014\)](#). These researchers used the same training procedure adopted in a verbal WM training procedure ([Borella et al., 2010](#)), based on the promising results achieved in terms of transfer effects and their maintenance, with visuospatial stimuli (although they did not compare directly verbal vs visuospatial modality within the same study). In particular, in the verbal version of the WM training (2010), the training task involved remembering the last word in a sequence of words and tapping the table whenever an animal noun was heard. Differently, in the visuospatial version (2014), the WM training task consisted in recalling the last position of a dot (maintenance phase) that was moved several times around a 4×4 matrix. Participants were also required to press the spacebar whenever a dot appeared in a grey cell (processing phase). Results from these two studies were very different: [Borella et al. \(2010\)](#), with the verbal WM procedure, showed both specific gains and transfer effects to short-term memory, inhibition, processing speed and fluid intelligence. These benefits were also maintained at follow-up and replicated in subsequent studies with different populations (e.g. in the old-old, see [Borella et al., 2013](#)). Differently, in the visuospatial version ([Borella et al., 2014](#)), a specific effect was obtained in the criterion task (and was the only effect to be maintained at follow-up), and only a near transfer effect to short-term memory. The authors attributed the larger transfer benefits related to the verbal WM training modality, to factors such as the greater age-related decline in the visuospatial component of WM (e.g., [Myerson, Emery, White, & Hale, 2003](#)). Furthermore, [Borella et al. \(2014\)](#) also suggested the need to take the particular type of stimuli used in the visuospatial procedure (dots) into account. Compared to words, dots are less meaningful to older adults (e.g. [Cornoldi, Bassani, Berto & Mammarella, 2007](#)). The WM visuospatial training task, contrary to the verbal one in which concrete words were used, may have thus limited participants' awareness in terms of both their improvement with respect to the stimuli correctly recalled and errors. Such different types of stimuli may have lowered engagement in activities to prevent memory errors and actively control for irrelevant information, an aspect that could be related to the lack of transfer effects. The use of dots that are always the same in all the training sessions, may have also produced a non-challenging training task, which in turn reduced participants' motivation and interest in the activities, all aspects that drive transfer effects.

The nature and the type of the WM training task stimuli thus deserve to be investigated and clarified. To this end, the aim of the present study was therefore to assess the efficacy of a visuospatial WM

training (using the same procedure as in [Borella et al., 2014](#)) in which we varied the type of visual stimuli used.

In order to overcome some of the aforementioned limits of the study by [Borella et al. \(2014\)](#) linked to the use of dots, in the present study, we used, images, instead of dots, in the training tasks in two experiments. Images are more meaningful than dots and easier to identify and therefore may attract more attentional resources. Stronger recruitment of attentional resources and enhancement by the training may promote training transfer effects. At the same time, using different images at each set may also make tasks more challenging than using dots and foster participants' motivation in performing the tasks and result in stronger cognitive commitment. All these aspects could thereby improve both specific and WM training transfer effects.

To this end, we used neutral words in the first experiment. In the second experiment (experiment 2), run simultaneously, we speculated that positive images might be even more salient than neutral stimuli in the case of older adults (e.g., [Mather, 2007](#)). Therefore, while maintaining the same training procedure as in experiment 1, we used positive stimuli instead of neutral words in line with evidence about the “positivity effect” in aging, that is the tendency of older adults to detect and remember positive information better than neutral or negative information. According to socioemotional selectivity theory (for a review see [Mather & Carstensen, 2005](#)), this effect seems to influence both memory and attentional tasks, and is due to changes in emotional regulation patterns in aging. Other studies have also reported a positivity effect on WM performance showing how older adults perform as well as younger adults when positive stimuli are involved (e.g., [Mikels, Larkin, Reuter-Lorenz, & Carstensen, 2005](#); [Mammarella, Fairfield, Frisullo, & Di Domenico, 2013](#)). Therefore, positive stimuli may increase recruitment of attentional resources and well-preserved emotional stimuli processing may compensate for mechanisms that decline with aging, such as WM ([Dolcos, LaBar, & Cabeza, 2004](#); [Mammarella et al., 2013](#); [Fairfield, Mammarella & Di Domenico, 2015](#)) and visuospatial WM, which is more sensitive to aging (e.g., [Myerson et al., 2003](#)). To the best of our knowledge, no study has yet investigated this possibility in older adults and the few studies that have examined the role of positive information in WM training have focused on younger adults alone (e.g., [Schweizer, Hampshire, & Dalgleish, 2011](#)). Furthermore, the specific aim of such studies was to see if WM training could improve emotional regulation, given that emotion regulation and WM largely overlap in the frontoparietal network (e.g., [Ochsner & Gross, 2005](#)). In experiment 2, we thus hypothesized that specific gains and related transfer effects might be fostered in older adults by using positive material in the training tasks.

In both experiments, efficacy was assessed in terms of specific gains in visuospatial WM and transfer effects to verbal WM, short-term memory, processing speed and fluid intelligence. Maintenance effects were also considered at an 8-month follow-up.

In the light of the study by [Borella et al. \(2014\)](#), we expected to see specific gains, and to find them maintained at follow-up. We also hypothesized that transfer benefits might be generated by positive stimuli, especially in experiment 2. These latter, due to maintained emotional regulation processes in aging (e.g., [Mather & Carstensen, 2005](#)), might sustain cognitive flexibility and compensate for difficulties encountered in the WM training activities.

These effects were expected to emerge when comparing the training group with an active control group.

2. Experiment 1

2.1. Method

2.1.1. Participants

Thirty-five 63- to 73-year-old adults took part in the study and were randomly assigned to trained ($n = 18$) or active control ($n = 17$) groups. All participants were healthy, native Italian speakers, lived independently, and volunteered for the study. All participants were selected on the grounds of a physical and mental health questionnaire. None met the exclusion criteria suggested by [Crook et al. \(1986\)](#). At baseline, all participants scored 9 or 10 on the short version of the Italian checklist for the Multidimensional assessment (SVAMA) for older adults adopted in the Veneto region ([Gallina](#)

Table 1

Experiment 1: demographic characteristics of the training and control groups.

	Training group (n = 18; 10 F)		Control group (n = 17; 11 F)	
	M	SD	M	SD
Age	68.11	3.16	68.82	2.92
Years of education	11.50	2.64	12.05	2.51
Vocabulary	48.22	11.19	46.47	10.39

M: mean; SD: standard deviation; F: female.

et al., 2006), indicating that they had good cognitive capacities. None scored below the norm in the vocabulary test of the Wechsler Adult Intelligence Scale, Revised (WAIS-R; Wechsler, 1981; Italian norms by Orsini & Laicardi, 2003), and they all performed above cut-off their age and education in the verbal WM task (Categorization working memory task – see Italian norms, De Beni, Borella, Carretti, Marigo, & Nava, 2008).

As demonstrated by one-way analyses of variance (ANOVAs), the experimental and control groups did not differ in terms of age, years of education, or WAIS-R vocabulary scores (all $F_s < 1$, see Table 1).

2.2. Materials

2.2.1. Specific effects

2.2.1.1. Criterion task: the Matrix task (Cornoldi et al., 2007; see also Borella et al., 2014). This is a dual task that involves retaining and simultaneously processing visuospatial information. It is a computer-based task consisting of sixty 4×4 matrices, grouped into sets of different length and consequently increasing difficulty (from two to six). In each set, an image is shown three times in a different position in the matrix. At the end of each set, respondents are asked recall the position of the last image seen in the set of matrices (maintenance phase). Each matrix includes one (randomly selected) row and column shaded in grey, and respondents are asked to press the spacebar whenever they see an image in a grey cell (processing phase). We created a parallel version by changing the images that had to be processed. Parallel versions were counterbalanced across testing sessions. The respondent's WM capacity (maximum score 60) is indexed by the total number of correctly recalled image positions.

2.2.1.2. Categorization working memory span task (CWMS; De Beni et al., 2008). The task consists of 40 word lists, divided into sets containing from 2 to 6 lists each. Participants listen to a set of word lists audio-recorded at a rate of 1 s per word, and they tap on the table with their hand whenever they hear an animal noun (processing phase). At the end of a set of word lists, participants are asked to recall the last word on each list (maintenance phase), i.e. they need to remember from 2 to 6 words altogether, depending on the difficulty of the set. Half (twelve) of the sets were used for the pretest session, the other 12 at the post-test session, and they were counterbalanced across testing sessions.

The total number of correctly recalled words was used as the measure of WM performance (maximum score 20).

2.2.1.3. Forward and backward Corsi span (adapted from the Corsi Blocks Task, 1972). This task consists of nine blocks placed randomly on a wooden board. The experimenter sequentially taps a number of blocks at a rate of one block per second, and participants reproduce the sequence in the same order (Forward condition) or in reverse order (Backward condition). The level of difficulty gradually increases from a minimum of 3 blocks to a maximum of 8 in the Forward condition, and from a minimum of 2 to a maximum of 7 in the Backward condition. Two sequences of the same length are presented for each level of difficulty, and the task is suspended after two consecutive recall errors. We created two versions of each task by exchanging the digit strings within each level of difficulty. We used one version for pretest the other for post-test. Versions were counterbalanced across testing sessions. The total number of correctly recalled sequences was considered as the final score (maximum score 12, for both tasks).

2.2.1.4. Pattern comparison task (adapted from [Salthouse & Babcock, 1991](#)). This task consists of two pages, each containing 30 items consisting of arrangements of lines and segments, located on the left and right of the page. Participants are asked to decide as quickly as possible whether or not the stimuli presented on the two sides are identical. The experimenter uses a stopwatch to record the time it takes to complete each page.

We created two versions of the task by inverting page order. We administered one version at pretest and the other at post-test. Versions were counterbalanced across testing sessions. The dependent variable was the total time (in sec) taken to complete the two pages.

2.2.1.5. Culture Fair Test, scale 3 ([Cattell & Cattell, 1963](#)). Scale 3 of the Cattell test consists of 4 subtests to be completed each within 2.5 to 4 min (depending on the subtest). In each subtest, participants are asked to:

- complete an incomplete series of figures, choosing which of 6 options best completed the series;
- identify figures or shapes that differed from the others in a series;
- choose items that correctly completed matrices of abstract figures and shapes;
- assess the relationship between a series of items.

We administered one of the two parallel forms (A or B) pretest and the other at post-test, in a counterbalanced fashion across testing sessions. The dependent variable was the number of correctly answers items across the four subsets (maximum score 50).

Parallel versions of each proposed tasks were used in order to avoid practice effects driven by item familiarity ([Salthouse, Schroeder, & Ferrer, 2004](#)).

2.3. Procedure

Participants in the training and control groups all attended five individual sessions: the first and last sessions for pre- and post-test assessments (see [Table 2](#) for the task presentation order), while the training or control activities were administered during the other 3 sessions, held within a 2-week time frame, maintaining a 2-day break between one session and the next. The schedule and duration of the sessions were the same for both groups in order to match the quantity of social interaction with the experimenter conducting the training. The duration of all session was about 60 minutes consistent with previous studies using the same procedure ([Borella et al., 2014](#)).

Participants in the training group were trained using modified versions of the Matrix task (the criterion task). At each training session, this task varied in terms of input retention and processing requirements (see [Table 2](#) for the training and control groups' activities).

2.4. Data analyses

Preliminary one-way ANOVAs were conducted for each of the measures used in the assessment to identify any baseline (pre-test) differences between the training and control groups. Then, to examine training-related gains, a 2- (group: trained, control) $\times$ 3 (session: pre-test, post-test, follow-up) mixed-design ANOVA was run for all measures of interest. Significant main effects and interactions were analyzed using pairwise comparisons, with Bonferroni's adjustment for multiple comparisons. The α value was set at .05 for all statistical tests, and at .006 for interactions (because nine comparisons were conducted [$0.05/9 = 0.006$]).

2.5. Results

The one-way ANOVAs revealed no significant differences between the training and control groups in any measure at the pre-test session. For all measures, $F < 1$, except for the CWMS task, $F_{(1,34)} = 1.07$, $P = .30$, and the Forward Corsi Span task, $F_{(1,34)} = 1.15$, $P = .29$ ([Table 3](#)).

The results of 2×3 ANOVAs (values of F , P , MSE and ηp^2) are summarized in [Table 4](#).

Table 2
Structure of training and control activities.

Session	Training group	Control group
1. Pre-test	Health interview, SVAMA, vocabulary test, forward and backward Corsi tasks, pattern comparison test, CWMS task, Image matrix task, Cattell test	
2. Training	WM training: increasingly long series comprising from two to five sets of 4 × 4 matrices, presented one after the other. Participants had to remember the positions of the last image presented and press the spacebar whenever an image occupied a gray cell. The WM training task included three phases, completed sequentially for each level of difficulty (or length of the series of images): in phase 1, participants had to recall the position of the last image in each series of matrices; in phase 2, they recalled the position of the first image in each series of matrices; and in phase 3, they recalled the position of the last image again. In each phase, if the position of the image to recall was correctly remembered, the task's difficulty was increased, up to five sets of matrices. If a mistake was made in one of the three phases, participants were presented with the next set of matrices, starting from the easiest level	Autobiographic memory questionnaire (De Beni et al., 2008)
3. Training	WM training: four sets of matrices for each different length (from two to five). The complexity of the task was manipulated by reducing or increasing the number of gray cells. For each matrix, participants had to press the spacebar whenever an image occupied a gray cell, as well as remembering the position of the last image displayed in each matrix	Memory sensitivity (De Beni et al., 2008)
4. Training	WM training: four sets of different difficulty (involving from two to five positions to recall). Participants were asked to press the spacebar whenever an image occupied a gray cell, and they had to recall the last positions displayed in the first set (a); the first positions in the second (b); the last positions in the third (c); and the first positions in the fourth (d)	Psychological well-being (De Beni et al., 2008)
5. Post-test	Forward and backward Corsi tasks, Pattern comparison test, CWMS task, Matrix task, Cattell test	

Adapted from Borella et al., 2014.

Table 3
Experiment 1: descriptive statistics for the measures of interest by group (training and control).

Variable	Training group (n = 18)						Control group (n = 17)					
	Pre-test		Post-test		Follow-up		Pre-test		Post-test		Follow-up	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
Image matrix task Max = 60	30.22	4.79	37.66	7.94	39.38	6.26	32.94	5.70	34.64	6.95	33.05	6.31
CWMS Max = 20	10.55	4.18	11.44	3.55	11.27	4.30	9.29	2.84	9.52	3.59	9.17	3.72
Forward Corsi span Max = 12	5.22	1.35	5.83	1.20	5.44	1.33	4.76	1.14	5.29	1.86	5.47	1.87
Backward Corsi span Max = 12	6	1.81	6.16	1.91	5.94	1.39	5.64	1.83	6.29	1.99	5.52	1.17
Pattern comparison (RT)	164.66	42.08	145.94	42.42	157.61	35.68	163.23	43.31	145.05	43.77	150.88	38.79
Cattell test Max = 50	16.77	3.99	18.83	4.09	17.44	2.93	17.52	4.51	18.05	4.89	16.11	5.07

M: mean; SD: standard deviation; RT: reaction time (in sec).

Table 4
Experiment 1: mixed-design 2 × 3 Anova results for the measures of interest, with group (training and control) as between-subjects factor, and sessions (pre-test, post-test, and follow-up) as repeated measures.

		$F_{(2,66)}$	MSE	P	ηp^2
<i>Criterion task: specific effect</i>					
Image matrix task	G	1.26	100.98	.26	.03
	S	22.36	11.07	.00**	.40
	GxS	16.55	11.07	.00**	.33
<i>Nearest transfer effect</i>					
CWMS	G	2.50	32.38	.12	.07
	S	.54	4.82	.56	.01
	GxS	.35	4.82	.70	.01
<i>Near transfer effect</i>					
Forward Corsi span	G	.68	4	.41	.02
	S	2.45	1.31	.10	.06
	GxS	.62	1.31	.54	.01
Backward Corsi span	G	.19	6.14	.66	.00
	S	1.81	1.34	.17	.05
	GxS	.57	1.34	.56	.01
<i>Far transfer effects</i>					
Pattern comparison task	G	.05	4333.45	.81	.00
	S	8.15	365.22	.00**	.19
	GxS	.24	365.22	.78	.00
Cattell test	G	.13	38.38	.71	.00
	S	3.16	8.42	.04*	.08
	GxS	1.20	8.42	.30	.03

MSE: mean standard error; G: group; S: session; GxS: group × session Interaction.

* $P < .05$.

** $P < .001$.

2.5.1. Criterion task: the Matrix task

The main effect of group was not significant. However, the effect of session was significant, all participants recalled more positions at post-test than at pre-test ($P < .001$), and at follow-up compared to pre-test ($P < .001$). There were no differences between post-test and follow-up ($P = 1$). The Group × Session interaction was also significant. Post-hoc comparison showed that trained participants performed better at post-test than at pre-test ($P < .001$), and at follow-up compared to pre-test ($P < .001$), while there were no significant differences between performance at post-test and follow-up ($P = .21$). The control group showed no significant improvement in performance from pre-test to post-test ($P = .61$), or from pre-test to follow-up ($P = 1$). There was also no significant difference between post-test and follow-up ($P = .31$). The training group outperformed the control group at follow-up ($P < .01$).

2.5.1.1. Transfer effect. No transfer effects were found (Table 4).

2.6. Discussion

The aim of this first experiment was to assess the effectiveness of a visuospatial WM training procedure in terms of specific gains and transfer benefits to verbal WM, short-term memory, processing speed and fluid intelligence. Adopting the same visuospatial procedure as that in Borella et al. (2014), we wanted to see whether transfer effects could be fostered by manipulating the stimuli of the WM training task. We assumed that images with a neutral valence could be more meaningful for older adults than the dots used in the earlier study (Borella et al., 2014):

- fostering, as aforementioned, attentional resources toward the task;
- making the tasks more “interesting” and challenging – aspects that can improve their motivation during the training activities.

All images were categorizable with a verbal label, an aspect that we thought could foster the involvement of active control of irrelevant information (Cornoldi et al., 2007).

Consistent with the literature on WM training, we found gains in the WM criterion task confirming the possibility of improving WM performance in older adults. These specific gains were also maintained at the 8-month follow-up only in the training group. Contrary to our hypotheses, we found no significant transfer effects.

Unlike the study of 2014, we did not find any nearest transfer effects to verbal WM, even though the demographic characteristics of the training and control groups in the present and previous study (Borella et al., 2014) were comparable, as were training procedures and tasks (apart from the manipulation of stimuli, from dots to images). Although this training procedure might improve the capacity to process information in WM – as shown by the specific training gains –, it did not improve the more general cognitive flexibility. In other words, the use of images, instead of dots, did not foster the transfer of training gains.

A possible explanation for the present results therefore lies in that meaningful images did not produce, as expected, higher recruitment of attentional resources. It seems that this type of stimuli favored poorer allocation of resources pushing participants to spend more attentional resources on processing the stimuli *per se*, instead of focusing on retaining their spatial locations and making a more flexible use of their own resources. Such a result seems to be due to the fact that older adults have difficulty with visuospatial tasks that involved binding objects to locations (e.g., Cowan, Naveh-Benjamin, Kilb, & Sauls, 2006). Those difficulties might also become especially evident when contextual or irrelevant visuospatial information is involved (Mammarella, Fairfield, De Beni, & Cornoldi, 2009).

It is also possible that visuospatial WM training, because of the greater age-related decline in the visuospatial component of WM (e.g., Myerson et al., 2003), needs more training sessions to be effective, however. It is worth to mention that some authors have suggested a dose-dependent effects in WM training, indicating that a high number of training sessions might produce more benefits and transfer effects (e.g., von Bastian & Oberauer, 2013). As discussed below, it may also be that our participants developed and used a particular strategy that is task specific.

To conclude, the findings of experiment 1 suggest that older adults can benefit from WM training conducted using a visuospatial modality as showed by specific gains in WM task, but there were not transfer effects. In our second experiment, run at the same time as experiment 1, we further manipulated the type of stimuli, presenting positive images. Our hypotheses were the same as in experiment 1. We expected cognitive resources to be more focused on the task and to promote transfer effects due to the greater involvement of attentional resources toward visual stimuli. We expected WM transfer effects to be fostered by the use of images instead of abstract dots, and even more so by the use of positive stimuli, given the known positivity effect associated with aging.

3. Study 2

3.1. Method

3.1.1. Participants

Forty healthy older adults (63–74 years old) were enrolled and randomly assigned to the training or control group. The inclusion criteria adopted and the demographic characteristics of the sample were the same as experiment 1. Four participants in the training group and 1 in the control group dropped out due to medical issues, so the final sample consisted of 35 individuals (16 in the training group and 19 in the active control group).

One-way ANOVAs showed that the experimental and control groups did not differ in terms of age, years of formal education, or WAIS-R vocabulary scores (all $F_s < 1$, see Table 5).

3.2. Materials

3.2.1. Criterion task: the Matrix task

This task was the same as in experiment 1 with the sole difference that the stimuli used here had a positive emotional valence.

Table 5
Experiment 2: demographic characteristics of the training and control groups.

	Training group (n = 16; 13 F)		Control group (n = 19; 15 F)	
	M	SD	M	SD
Age	68.18	3.63	69.31	3.23
Years of education	11.31	3.15	11.15	2.96
Vocabulary	50.43	6.99	48.42	8.40

M: mean; SD: standard deviation; F: female.

3.2.2. Transfer task

We used, in the same order, the tasks from experiment 1. We adopted parallel versions of each proposed task to avoid any test-retest effects.

3.3. Procedure

The pre- and post-test assessments were conducted in the same way as for experiment 1. The training schedule was also the same (Table 2), with the sole exception that emotionally positive stimuli were used for the training task.

3.4. Results

One way ANOVAs revealed no significant differences between the training and control groups at the pre-test, in any of the measures (all $F_s < 1$, see Table 6). The results of the 2×3 ANOVAs (in terms of F , p , MSE and ηp^2) are summarized in Table 7.

3.4.1. Criterion task: the Matrix task

The main effect of group was significant, and so was the main effect of session. The trained participants correctly recalled more image positions than controls ($P = .006$), and all participants' performance improved from pre-test to post-test ($P < .001$), and from pre-test to follow-up ($P < .001$), whereas their performance deteriorated from post-test to follow-up ($P < .001$). The Group $\times$ Session interaction was also significant. Post-hoc comparison showed that trained participants performed better at post-test

Table 6
Experiment 2: descriptive statistics for the measures of interest by group (training and control).

Variables	Training group (n = 16)						Control group (n = 19)					
	Pre-test		Post-test		Follow-up		Pre-test		Post-test		Follow-up	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
Image matrix task Max = 60	35.68	4.19	44.12	4.19	40.81	3.63	35.21	5.73	37.47	4.81	35.21	6.12
CWMS Max = 20	12.68	2.49	13.81	3.35	14.56	2.12	12	2.94	12.52	2.69	12.57	2.47
Forward Corsi span Max = 12	5.18	1.04	5.56	.89	5.75	1.06	4.31	1.05	4.42	1.07	4.68	1.29
Backward Corsi span Max = 12	5.31	1.01	5.62	1.08	6	1.09	5.42	1.64	5.42	1.53	5.57	1.16
Pattern comparison (RT)	143.31	30.66	138.68	18.22	136.12	21.27	153.68	36.24	144.57	25.55	140.88	23.48
Cattell test Max = 50	20.06	2.56	21.87	2.52	21.75	2.14	19.52	3.43	20.31	2.53	20.15	2.93

M: mean; SD: standard deviation; RT: reaction time (in sec).

Table 7

Experiment 2: mixed-design 2 × 3 Anova results for the measures of interest, with group (training and control) as the between-subjects factor, and sessions (pre-test, post-test, and follow-up) as repeated measures.

		$F_{(2,66)}$	MSE	P	np^2
<i>Criterion task: specific effect</i>					
Image matrix task	G	8.55	54.84	.00**	.20
	S	27.42	9.07	.00***	.45
	GxS	10.44	9.07	.00***	.24
<i>Nearest transfer effect</i>					
CWMS	G	2.67	16.97	.11	.07
	S	5.37	2.52	.00**	.14
	GxS	1.44	2.52	.24	.04
<i>Near transfer effect</i>					
Forward Corsi span	G	9.89	2.77	.00**	.23
	S	5.03	.37	.00	.13
	GxS	.45	.37	.64	.01
Backward Corsi span	G	.18	4.17	.67	.00
	S	3.58	.44	.03*	.09
	GxS	1.39	.44	.25	.04
<i>Far transfer effects</i>					
Pattern comparison task	G	.74	1704.07	.39	.02
	S	4.01	226.83	.02*	.10
	GxS	.34	226.83	.71	.01
Cattell test	G	2.62	15.02	.11	.07
	S	4.65	3.80	.01*	.12
	GxS	.82	3.80	.44	.02

MSE: mean standard error; G: group; S: session; GxS: group × session interaction.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

than at pre-test ($P < .001$), and at follow-up than at pre-test ($P < .001$); the trained group's performance was significantly worse at follow-up than at post-test ($P < .01$). The control group showed no significant differences between the sessions. The training group performed significantly better than the control group at post-test ($P < .001$), and at follow-up ($P < .01$).

3.4.1.1. *Transfer effect.* No significant transfer effects were found (Table 7).

3.5. Discussion

The aim of this second experiment was to assess, in older adults, the efficacy of a visuospatial WM training with emotionally positive stimuli. As in the previous experiment, we assumed that the type of stimuli used in the training task could have a bearing on the generation of transfer effects, and that emotionally positive stimuli could foster these effects due the well-known “positivity effect” in aging.

The significant improvement found in the trained group for the criterion task goes to show that this type of visuospatial procedure can improve visuospatial WM performance in older adults, when emotionally positive stimuli are used (this specific effect was also maintained at follow-up). Contrary to our expectations, however, and consistent with results obtained in experiment 1, there was no sign of any transfer effects to verbal WM, short-term memory, processing speed, or fluid intelligence.

As in experiment 1, it may be that the particular tasks used (which involved sequentially recalling the position of images on a matrix) place an excessive additional burden on the mental resources of older adults rather than promoting plasticity. It may also be that the improvements found in both experiments are linked to having learned a strategy that benefited very similar tasks (specific training gain) (Li et al., 2008). Participants, contrary to our hypotheses, might have allocated more mental resources to the stimuli presented *per se*. This might be even more important when emotionally

positive stimuli are presented because, alongside reports that positive stimuli can compensate for difficulties in WM tasks (Mammarella et al., 2013), others have shown that their processing might subtract resources from the analysis of non-emotional information (Mather, 2007).

4. General discussion

In two experiments, we aimed to clarify whether the type of stimuli used in a visuospatial WM training procedure could influence the efficacy of the training in terms of specific gains and transfer effects in older adults.

Whether we used neutral or emotionally positive images as stimuli in the WM training task, our results showed that the visuospatial training procedure used here generated specific benefits in the criterion task. The presence of such specific training gains only in the training group is in line with WM training studies in aging see Karback & Verhaeghen, 2014, and the use of parallel versions, indicate that they are more plausibly attributable to the WM training rather than item-specific practice.

On the other hand, no transfer effects were shown. The present findings, although in contrast with the results of the verbal WM training of Borella et al. (2010) in which even far transfer effects occur, are, however, generally consistent with those of the previous study conducted by Borella et al. (2014) using visuospatial stimuli (dots). Therefore, the present results, along with those of Borella et al. (2014), seem to suggest that when using visuospatial WM training tasks, the type of stimuli has only a marginal role in contributing to any transfer effects. The use of images (neutral or emotionally positive ones), instead of meaningless stimuli (dots), did not favor transfer effects – not even nearest transfer effects to a verbal WM (as seen in Borella et al., 2014). Several explanations can account for the actual lack of transfer effects in present results:

- visuospatial WM is more liable to age-related decline than verbal WM (e.g. Myerson et al., 2003). It may be that verbal WM training induces transfer effects in older people more easily than visuospatial training because it focuses on abilities that are better preserved. According to some authors, training programs could help older adult brains to build a compensatory scaffold in response to age-related changes – leading them to focus on abilities that are more age-resilient – and on this compensatory scaffold they could establish a greater cognitive flexibility (Park & Reuter-Lorenz, 2009; Borella et al., 2014). This might be the case of verbal abilities, rather than the visuospatial ones, as they more closely related to crystallized abilities, and therefore to previously learned knowledge and experiences that are maintained in aging (e.g., Haavisto & Lehto, 2005);
- the type of stimuli used may have focused resources on the processing of the images *per se*, rather than eliciting greater attentional resources on general stimuli processing in light of a more flexible use of the resources. Our participants were probably not helped, but distracted by the characteristics of the images; because of this phenomenon the demands of the criterion task used in this study on our older participants' cognitive resources may have consequently been too great for them to experience transfer benefits;
- a third interpretation is related to the training length. Indeed, although the brevity of the Borella et al. (2010) training (the verbal procedure) represents a strength (especially in clinical setting), since the type of visuospatial materials used and the accelerate decline of visuospatial WM, more training sessions could be helpful to train the visuospatial component of WM and detect transfer effects in older adults. It is worth adding, however, that the mean number of positions correctly recalled in the criterion task, before and after the training, in the two present experiments was even higher than in the study conducted in 2014. Looking at the data in qualitative terms, Cohen's *d* values for the effect sizes were comparable ($d = 2.25$ in the previous study by Borella et al. [2014]; $d = 1.40$ in the present experiment 1; and $d = 2.21$ in experiment 2), and in all three cases, there were large effects (higher than .80) in the criterion task, indicating robust specific effects that were maintained at follow-up. Despite these large specific effects, there was no evidence of any transfer effects in the present study, in both experiments;
- it may be thus that specific strategies were developed to deal with the visuospatial WM tasks requests (see Borella et al., under review). The enhanced efficiency in visuospatial WM task identified could

thus be mediated by strategy use. In other words, participants may have acquired a strategy suited to the tasks used during the training and then applied it to the criterion task. Improvements due to strategy use are generally task-specific and lead to only a narrow effect (von Bastian & Oberauer, 2014). However, we have no information concerning strategy use, including the type of strategy, by our participants. Future studies should thus investigate this issue more directly, by assessing the strategy used.

Overall, the results of this study indicate that unlike some studies showing a positive relationship between amount of specific gains and transfer benefits (e.g., von Bastian & Oberauer, 2013; Zinke et al., 2014), our training procedure generated only a large specific effect in the criterion task. The present pattern of findings also suggests the type of stimuli used in a visuospatial WM training task may have only a marginal influence in inducing any transfer effects. The task modality *per se* (verbal vs visuospatial) seems to be more influential in this sense. Such an hypothesis warrants further investigation.

This work has several important limitations. The first relates to the small size of our sample: although it is comparable with the one used in previous studies on WM training in older adults. Results should be replicated in larger samples. Second, it would be interesting to compare the effects of different kinds of visuospatial stimuli (dots vs neutral images or positive ones) within the same study. A further limitation of the present study lies in the absence of an incidental memory task relating to the number of images recalled at the end of the training task. Such a task could have shed light on the hypothesis that using images instead of dots meant that some cognitive resources were diverted to the characteristics of the stimuli *per se*, rather than focusing on their position.

This study should be replicated taking the rate at which the visuospatial stimuli were presented in the training task into account because the time allotted to complete a task may well influence WM performance (e.g. Salthouse, 1996; Nettelbeck & Burns, 2010), in particular when the task is the training one. This is even more important in older adults, considering the age-related decline in processing speed (Cerella & Hale, 1994), as well as in visuospatial ability (e.g. Myerson et al., 2003). Slowing the rate of presentation of the visuospatial stimuli (and, in particular, allowing more time across sequences of matrices to favor the retention of relevant information) generally leads to an improvement in older adults' performance. The actual rate of item presentation may be too fast for older adults (as reported by one of our participants) who may perceive the task too difficult and frustrating disrupting motivation toward the activities and limiting the occurrence of transfer effects. According to the model of Lövdén, Bäckman, Lindenberger, Schaefer, & Schmiedek (2010), cognitive plasticity may occur only if the mismatch between the individual's cognitive resources and task request is at an "optimal point". If the task is too difficult, as well as too easy, cognitive plasticity (that is the foundation of transfer effects) does not occur; if gains are present, they are only attributable to the fact participants simply learn how to ameliorate performance in the trained task by the use, for example, of content-specific strategies, the so-called cognitive flexibility. We hypothesize thus, that slowing the rate of presentation of stimuli, (as well as introducing more training sessions), may promote "cognitive flexibility" and foster "cognitive plasticity".

WM training studies would also be of sure interest to go on with this fascinating topic: future researches could directly compare the effects of verbal and visuospatial procedure with a within design, or investigate the effects of the verbal stimuli if the valence of words is manipulated. It is possible that emotional positive words if used in a verbal WM training procedure, differently from what was found here with visuospatial training modality, could effectively enhance WM training gains and transfer effects, due the well preserved verbal component of WM (e.g., Myerson et al., 2003) combined with the positivity effect (Mather & Carstensen, 2005).

Overall the two experiments add to what we know about the influence of WM task modality (verbal versus visuospatial), and the stimuli involved, in triggering transfer benefits in WM training. Our findings using the present visuospatial WM training paradigm suggest that it is less effective, in terms of transfer effects, than the same paradigm administered verbally in a previous study, regardless of the type of stimuli used in WM training tasks. Also, and in a completely novel way, we shed light on the role of positive stimuli in WM training for older adults.

Disclosure of interest

The authors declare that they have no competing interest.

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